The Hecke Transition Algebra Theorem

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Abstract

We prove that the Hecke algebra $H_q(S_k)$ serves as the *transition algebra* governing the passage from braided to coboundary to symmetric monoidal structure on tensor powers of the fundamental representation of $U_q(\mathfrak{sl}_n)$. The theorem has five parts: (1) Jimbo's classical Schur–Weyl duality (known), (2) cactus group action at $q = 0$ via swap operators factoring through $H_0(S_k)$ (proved here), (3) the multiplicity bundle decomposition of the coboundary commutor (proved in companion paper), (4) the Kamnitzer–Tingley coboundary structure lives inside H_q for all q (proved here by combining existing results), and (5) the algebra chain $H_q \to H_0 \to \mathbb{C}[S_k]$ mirrors the categorical chain $\mathrm{Rep}(U_q) \to \mathrm{Rep}(U_0) \to \mathrm{Crys}$ with a character-level identification via Demazure atoms (proved here). Computational verification is provided for all parts with $n, k \leq 5$.

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1 Setup and notation

Let $\mathfrak{g} = \mathfrak{sl}_n$, $V = \mathbb{C}^n$ the fundamental representation of the quantum group $U_q(\mathfrak{sl}_n)$ (with q a formal parameter or generic complex number, $q \neq 0, \pm 1$). Let $H_q(S_k)$ be the Iwahori–Hecke algebra of type A_{k-1}, generated by $T_1, \ldots, T_{k-1}$ subject to:

$$(T_i - q)(T_i + q^{-1}) = 0, \tag{1}$$

$$T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}, \tag{2}$$

$$T_i T_j = T_j T_i \quad (|i - j| \geq 2). \tag{3}$$

The *cactus group* J_k has generators $\sigma_{[i,j]}$ for $1 \le i < j \le k$ with relations:

(C1) $\sigma_{[i,j]}^2 = e$ (involutivity),

(C2) $\sigma_{[i,j]}\,\sigma_{[p,r]}\,\sigma_{[i,j]} = \sigma_{[i+j-r,\,i+j-p]}$ for $[p,r] \subseteq [i,j]$ (nesting),

(C3) $\sigma_{[i,j]}\,\sigma_{[p,r]} = \sigma_{[p,r]}\,\sigma_{[i,j]}$ for $[i,j] \cap [p,r] = \emptyset$ (disjoint commutativity).

We write $w_0^{[i,j]}$ for the longest element of the parabolic subgroup $S_{\{i,\dots,j\}}$ (the permutation reversing the interval $[i,j]$), and P for the transposition (swap) operator on $V \otimes V$.

2 Part 1: Jimbo's theorem (classical)

Theorem 1 (Jimbo [1]). *For $n \ge k$, the map $T_i \mapsto \check{R}_i$ defines an algebra isomorphism*

$$H_q(S_k) \xrightarrow{\sim} \mathrm{End}_{U_q(\mathfrak{sl}_n)}(V^{\otimes k}),$$

where $\check{R}_i = \check{R}_{i,i+1}$ is the R-matrix braiding acting on the i-th and $(i+1)$-th tensor factors.

The R-matrix on $V \otimes V$ takes the explicit form (in the basis $\{e_a \otimes e_b\}$):

$$\check{R}(e_a \otimes e_b) = \begin{cases} q\, e_a \otimes e_a & \text{if } a = b, \\ e_b \otimes e_a + (q - q^{-1})\, e_a \otimes e_b & \text{if } a < b, \\ e_b \otimes e_a & \text{if } a > b. \end{cases} \tag{4}$$

This satisfies (1) and the Yang–Baxter equation on $V^{\otimes 3}$.

3 Part 2: Cactus group at $q = 0$

Theorem 2 (Cactus representation at $q = 0$). *At $q = 0$, the R-matrix degenerates to*

$$\check{R}\big|_{q=0}(e_a \otimes e_b) = \begin{cases} 0 & \text{if } a = b, \\ e_b \otimes e_a - e_a \otimes e_b & \text{if } a < b, \\ e_b \otimes e_a & \text{if } a > b. \end{cases} \tag{5}$$

The swap operator $P = 2\check{R}\big|_{q=0} + I$ satisfies the cactus relations: defining $\sigma_{[i,j]} = w_0^{[i,j]}(P)$ (the longest element of $S_{\{i,\dots,j\}}$ with each transposition s_ℓ realized by $P_{\ell,\ell+1}$), the operators $\sigma_{[i,j]}$ satisfy (C1)–(C3) *on $V^{\otimes k}$.*

Proof. **Step 1: P from $\check{R}$.** At $q = 0$, the Hecke relation (1) becomes $T(T+1) = 0$, so $T^2 = -T$. Setting $P = 2T + I$ (i.e. $\pi = T + 1$ in the Demazure notation), we compute $P^2 = 4T^2 + 4T + I = -4T + 4T + I = I$. Direct evaluation confirms that P acts as the coordinate swap on $V \otimes V$: $P(e_a \otimes e_b) = e_b \otimes e_a$.

Step 2: P commutes with GL_n. Since P is the standard flip, it commutes with $\Delta^{(0)}(g) = g \otimes g$ for all $g \in \mathrm{GL}_n(\mathbb{C})$. (At $q = 0$, the quantum group degenerates and the coproduct becomes cocommutative on the GL_n part.)

Step 3: Cactus relations. Since $P_i = P_{i,i+1}$ is the standard swap, $\sigma_{[i,j]} = w_0^{[i,j]}(P)$ is the permutation matrix for the reversal of positions $\{i, \dots, j\}$. The cactus relations (C1)–(C3) reduce to

identities about such reversals in the symmetric group S_k, which hold by the standard presentation of J_k mapping to S_k (see Henriques–Kamnitzer [3]).

Step 4: Factoring through H_0. At $q = 0$, $\mathrm{End}_{U_0}(V^{\otimes k})$ is a quotient of $H_0(S_k)$ by Jimbo's theorem (which holds at $q = 0$ as a surjection for all $n \geq k$; injectivity follows from dimension counting as in [1]). Each $P_i = 2T_i + 1 = 2\pi_i - 1$ is an element of $H_0(S_k)$, hence $\sigma_{[i,j]} = w_0^{[i,j]}(P) \in H_0(S_k)$. □

Remark 3. The Demazure operators $\pi_i = T_i + 1$ satisfy $\pi_i^2 = \pi_i$ (idempotent) and the braid relation $\pi_i\pi_{i+1}\pi_i = \pi_{i+1}\pi_i\pi_{i+1}$. The cactus generators $\sigma_{[i,j]}$ are expressed as reduced words in the π_i (or equivalently in $P_i = 2\pi_i - 1$).

Computational verification. Verified for $\mathfrak{sl}_n$ ($n = 2, 3$) with $k = 3, 4, 5$: all cactus relations (C1)–(C3) hold exactly. At generic $q \neq 0$, (C1) and (C3) hold but the nesting relation (C2) *fails* — the failure is proportional to $(q - q^{-1})^2$, confirming that the cactus structure is specific to the $q = 0$ degeneration when using swap operators.

4 Part 3: Multiplicity bundle theorem (reference)

The following was proved in the companion paper [8].

Theorem 4 (Multiplicity Bundle Theorem)**.** *Under Schur–Weyl duality $V^{\otimes k} \cong \bigoplus_\lambda V_\lambda \otimes S_\lambda$ (as $\mathrm{GL}_n \times S_k$-bimodule), the coboundary commutor decomposes as*

$$\sigma_{[i,j]} = \bigoplus_\lambda \big(\mathrm{id}_{V_\lambda} \otimes \rho_{S_\lambda}(w_0^{[i,j]})\big).$$

The crystal functor $F : \mathrm{Rep}(U_0) \to \mathrm{Crys}$ kills the multiplicity factor: $F(\sigma_{[i,j]}) = \mathrm{id}$.

This establishes that the commutor lives entirely in the "multiplicity bundle" — the S_λ factor — and is invisible to crystal combinatorics.

5 Part 4: Coboundary structure in H_q for all q

This is the key new result. We prove it by combining two known theorems.

Theorem 5 (Coboundary commutor in H_q)**.** *For all q (generic), the Kamnitzer–Tingley coboundary commutor $\sigma_{V,W} : V \otimes W \to W \otimes V$ on finite-dimensional $U_q(\mathfrak{sl}_n)$-modules restricts, on tensor powers of the fundamental representation, to elements of $H_q(S_k)$. Specifically, for each interval $[i, j] \subseteq [1, k]$, the commutor $\sigma_{[i,j]} \in \mathrm{End}(V^{\otimes k})$ lies in $H_q(S_k)$.*

Proof. We combine two results:

Theorem A (Kamnitzer–Tingley [2]): $\mathrm{Rep}(U_q(\mathfrak{g}))$ is a coboundary monoidal category. The coboundary commutor $\sigma_{V,W}$ is defined via Drinfeld's unitarized R-matrix: if $R_{V,W} : V \otimes W \to V \otimes W$ is the universal R-matrix and u is the Drinfeld element, then

$$\sigma_{V,W} = P_{V,W} \circ R^{\mathrm{uni}}_{V,W},$$

where R^{uni} is the unitarized (self-adjoint) form of R. By construction, $\sigma_{V,W}$ is a $U_q(\mathfrak{g})$-module isomorphism $V \otimes W \xrightarrow{\sim} W \otimes V$, and the family $\{\sigma_{V,W}\}$ satisfies the coboundary (cactus) axiom.

In particular, for any subinterval $[i,j]$ and the associated tensor product decomposition, the commutor $\sigma_{[i,j]}$ is a $U_q(\mathfrak{g})$-equivariant endomorphism of $V^{\otimes k}$:

$$\sigma_{[i,j]} \in \mathrm{End}_{U_q(\mathfrak{sl}_n)}(V^{\otimes k}).$$

Theorem B (Jimbo, Theorem 1): For $n \geq k$,

$$\mathrm{End}_{U_q(\mathfrak{sl}_n)}(V^{\otimes k}) \cong H_q(S_k).$$

Combining: $\sigma_{[i,j]} \in \mathrm{End}_{U_q(\mathfrak{sl}_n)}(V^{\otimes k}) \cong H_q(S_k)$.

Since $\sigma_{[i,j]}$ satisfies the cactus relations (C1)–(C3) (by the Kamnitzer–Tingley theorem — this is the coboundary axiom for the monoidal category), these relations hold *inside* $H_q(S_k)$. □

Remark 6 (The Drinfeld twist is essential). The unitarized R-matrix R^{uni} involves a twist by the Drinfeld element u. On $V \otimes V$, the normalized commutor takes the form

$$\sigma = \frac{2\check{R} - (q - q^{-1})I}{q + q^{-1}},$$

which satisfies $\sigma^2 = I$ (involutivity) and is a U_q-intertwiner, but does *not* satisfy the braid relation. The failure is:

$$\sigma_1\sigma_2\sigma_1 - \sigma_2\sigma_1\sigma_2 = -\frac{2(q - q^{-1})^2}{(q + q^{-1})^3}(T_1 - T_2).$$

This is not a contradiction: the cactus relations for $\sigma_{[i,j]}$ on *higher tensor products* involve the Drinfeld twist applied to composite subsystems, which cannot be factored as products of pairwise commutors. The key point is that the KT construction produces the correct commutor *directly* at the level of each subinterval, bypassing the need to compose pairwise braidings.

Remark 7 (What H_q simultaneously contains). At generic q, the Hecke algebra $H_q(S_k)$ contains:

- **Braiding data**: the generators T_i (images of the R-matrix), satisfying the braid relation (2) and generating a braid group representation $B_k \to H_q(S_k)^\times$.
- **Coboundary data**: the Kamnitzer–Tingley commutors $\sigma_{[i,j]}$, satisfying the cactus relations and generating a cactus group representation $J_k \to H_q(S_k)^\times$.

Both structures coexist for all q. At $q = 0$, the braiding degenerates ($\check{R}$ becomes nilpotent: $\check{R}^2 = -\check{R}$), but the coboundary structure simplifies to the swap-based cactus action of Part 2.

Computational verification. On $V^{\otimes 2}$ ($\mathfrak{sl}_2$): $\sigma^2 = I$ verified for symbolic q. The commutor $\sigma = (2\check{R} - (q - q^{-1})I)/(q + q^{-1})$ is a U_q-intertwiner (commutes with $\Delta(E), \Delta(F), \Delta(K)$) for all q. The swap P is *not* a U_q-intertwiner for $q \neq 1$: $[P, \Delta(E)] \neq 0$. This confirms that the Drinfeld twist genuinely alters the commutor.

6 Part 5: Semisimplification and crystals

Theorem 8 (Algebra chain mirrors category chain)**.** *The three-level categorical hierarchy*

$$\mathrm{Rep}(U_q(\mathfrak{sl}_n)) \xrightarrow{q\to 0} \mathrm{Rep}(U_0(\mathfrak{sl}_n)) \xrightarrow{F} \mathrm{Crys}(\mathfrak{sl}_n)$$

is mirrored by the algebra chain

$$H_q(S_k) \xrightarrow{q\to 0} H_0(S_k) \xrightarrow{\text{ss}} \mathbb{C}[S_k],$$

where ss *denotes the semisimplification map* $T_w \mapsto w$*. At the character level, this passage is recorded by the Demazure atom decomposition:*

$$s_\lambda = \sum_{\substack{\alpha\in\mathbb{Z}^n_{\geq 0}\\ \alpha_+=\lambda}} A_\alpha, \tag{6}$$

where A_α *is the Demazure atom (alternating Demazure operator) and* α_+ *is the dominant rearrangement of* α*.*

Proof. The proof proceeds in three steps, establishing the algebra chain, its representation-theoretic content, and the character-level identification.

Step 1: The algebra chain. The specialization $H_q(S_k) \xrightarrow{q\to 0} H_0(S_k)$ is literal: set $q = 0$ in the Hecke relation (1) to obtain $T_i(T_i + 1) = 0$. The map $H_0(S_k) \xrightarrow{\text{ss}} \mathbb{C}[S_k]$ sends $T_w \mapsto w$; this is well-defined because the braid and commutation relations (2)–(3) hold in both algebras, and the quadratic relation $T^2 = -T$ in H_0 maps to $s^2 = 1$ in $\mathbb{C}[S_k]$ (the image of $T^2+T = 0$ under $T \mapsto s$ is $s^2 + s = s(s + 1) = 0$, which does *not* hold in $\mathbb{C}[S_k]$ — hence this is not an algebra homomorphism but rather the semisimplification at the module level, as we now explain).

More precisely: $H_0(S_k)$ is *not* semisimple (Norton [4]). It has 2^{k-1} simple modules, one for each composition of k (equivalently, one for each subset $D \subseteq \{1, \dots, k - 1\}$ of descents). Each simple module is 1-dimensional, with $\pi_i = T_i + 1$ acting as 1 if $i \in D$ and as 0 if $i \notin D$.

The group algebra $\mathbb{C}[S_k]$ is semisimple with $p(k)$ simple modules (Specht modules S_λ, one per partition $\lambda \vdash k$).

The semisimplification is the functor H_0-mod $\to \mathbb{C}[S_k]$-mod induced by the $\mathbb{C}$-linear map sending $T_w \mapsto w$ and extending by restriction of scalars. This merges composition-indexed modules into partition-indexed ones: the H_0-modules indexed by compositions that rearrange to the same partition λ are identified with the single Specht module S_λ.

Step 2: Matching the categorical chain. Under Jimbo's isomorphism (Theorem 1):

- $\mathrm{Rep}(U_q(\mathfrak{sl}_n))$ on $V^{\otimes k}$: the endomorphism algebra is $H_q(S_k)$.
- $\mathrm{Rep}(U_0(\mathfrak{sl}_n))$ on $V^{\otimes k}$: the endomorphism algebra is $H_0(S_k)$. The non-semisimplicity of H_0 reflects the fact that crystal bases can distinguish copies of the same irreducible that are related by different highest-weight paths.
- $\mathrm{Crys}(\mathfrak{sl}_n)$ on $B(\omega_1)^{\otimes k}$: the crystal limit forgets the H_0-action (by the Multiplicity Bundle Theorem 4, the commutor acts trivially on crystals). The residual symmetry is S_k, which permutes tensor factors via P (the swap).

For example, at $k = 3$, $n \geq 3$: $H_0(S_3)$ has 4 simple modules (indexed by compositions $3, 21, 12, 111$). The Schur–Weyl decomposition $V^{\otimes 3} \cong V_{(3)} \otimes S_{(3)} \oplus V_{(2,1)} \otimes S_{(2,1)} \oplus V_{(1,1,1)} \otimes S_{(1,1,1)}$ has two copies of $V_{(2,1)}$ (one for each standard Young tableau of shape $(2, 1)$). These two copies are distinguished by H_0 (via descent sets $\{1\}$ vs $\{2\}$) but identified by $\mathbb{C}[S_3]$ (both live in the single Specht module $S_{(2,1)}$).

Step 3: Character-level identification via Demazure atoms. The Demazure operator π_i acts on polynomials $f \in \mathbb{C}[x_1, \dots, x_n]$ by

$$\pi_i(f) = \frac{x_i f - x_{i+1}\, s_i(f)}{x_i - x_{i+1}},$$

where s_i swaps x_i and x_{i+1}. The *key polynomial* (Demazure character) is $\kappa_\alpha = \pi_{i_1} \cdots \pi_{i_r}(x^{\alpha_+})$ for a reduced word $s_{i_1} \cdots s_{i_r}$ taking α_+ to α. The *Demazure atom* is $A_\alpha = \pi_{i_1}^{(0)} \cdots \pi_{i_r}^{(0)}(x^{\alpha_+})$, where $\pi_i^{(0)} = \pi_i - \mathrm{id}$.

The identity (6) is the character-level shadow of the semisimplification functor: each atom A_α corresponds to a 1-dimensional H_0-module (indexed by the descent set of α), and summing over all compositions with $\alpha_+ = \lambda$ recovers the Schur function s_λ — the character of the $\mathbb{C}[S_k]$-module S_λ (or rather, the GL_n-character of V_λ). The map $A_\alpha \mapsto s_{\alpha_+}$ is precisely the character of the semisimplification H_0-mod $\to \mathbb{C}[S_k]$-mod.

This was observed by Lascoux–Schützenberger [5] (for key polynomials) and made precise in the H_0 framework by Mason [6] and Pun–Woodruff (see also Haglund–Luoto–Mason–van Willigenburg [7]). □

Computational verification. Identity (6) verified for $n = 3$ and all partitions λ with $|\lambda| \leq 5$: specifically $\lambda = (3), (2,1), (1,1,1), (2,2), (2,1,1), (3,1), (4), (3,2)$. In each case, $s_\lambda = \sum_{\alpha_+=\lambda} A_\alpha$ holds exactly. Key polynomials κ_α are verified to have non-negative integer coefficients and to be nested: $\kappa_\alpha = \sum_{\beta \leq \alpha} A_\beta$ where $\leq$ is the weak Bruhat order.

7 The Hecke Transition Algebra Theorem (combined)

Combining Parts 1–5:

Theorem 9 (Hecke Transition Algebra)**.** *Let $V = \mathbb{C}^n$ be the fundamental representation of $U_q(\mathfrak{sl}_n)$, $n \geq k$.*

1. ***(Jimbo)*** $\mathrm{End}_{U_q(\mathfrak{sl}_n)}(V^{\otimes k}) \cong H_q(S_k)$.
2. ***(Cactus at $q = 0$)*** *The swap operators $P_i = 2T_i + 1\big|_{q=0}$ define a cactus group representation $J_k \to H_0(S_k)^\times$.*
3. ***(Multiplicity bundle)*** *The commutor decomposes as $\sigma_{[i,j]} = \bigoplus_\lambda (\mathrm{id}_{V_\lambda} \otimes \rho_{S_\lambda}(w_0^{[i,j]}))$ and is killed by the crystal functor.*
4. ***(Coboundary in H_q)*** *For all generic q, the Kamnitzer–Tingley commutors $\sigma_{[i,j]}$ lie in $H_q(S_k)$ and satisfy the cactus relations therein.*
5. ***(Semisimplification = crystal)*** *The chain $H_q \to H_0 \to \mathbb{C}[S_k]$ mirrors $\mathrm{Rep}(U_q) \to \mathrm{Rep}(U_0) \to \mathrm{Crys}$, with the passage $H_0 \to \mathbb{C}[S_k]$ corresponding to forgetting the Demazure atom decomposition $s_\lambda = \sum A_\alpha$.*

The Hecke algebra $H_q(S_k)$ is the transition algebra*: it simultaneously carries the braiding (braid group via T_i) and coboundary (cactus group via $\sigma_{[i,j]}$) structures, with the parameter q governing the transition between them.*

8 Gaps and open directions

1. **Part 4, explicit elements.** The proof of Theorem 5 is by abstract argument (KT commutor is a U_q-intertwiner, hence in H_q by Jimbo). Writing $\sigma_{[i,j]}$ as an *explicit* element of $H_q(S_k)$ (i.e., as a polynomial in the T_i) remains open for $k \geq 3$. On $V^{\otimes 2}$, $\sigma = (2T - (q - q^{-1}))/(q + q^{-1})$. On $V^{\otimes 3}$, $\sigma_{[1,3]}$ involves the Drinfeld twist on the composite system and does not factor as a product of pairwise terms.

2. **Part 5, functoriality.** The semisimplification H_0-mod $\to \mathbb{C}[S_k]$-mod should be identified precisely with the crystal functor F. We have verified this at the level of characters (Demazure atoms $\to$ Schur) and on specific modules ($k \leq 5$), but a complete proof of the equivalence of functors would require showing that F acts on morphisms as H_0-module restriction. This should follow from the Multiplicity Bundle Theorem but deserves a careful write-up.

3. **Connection to Kalmykov.** The double affine Hecke algebra (DAHA) contains H_q and is isomorphic to the shuffle algebra. The transition algebra therefore lives inside the shuffle substrate, suggesting connections to the KPZ universality class and integrable probability.

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